

Foundations of Autonomous vehicle

[Autonomous vehicle]

1.0 Content Block

Path planning Path planning finds a sequence of segments that a vehicle can use to move from origin to destination. Techniques used for path planning include graph-based search and variational-based optimization techniques. Graph-based techniques can make harder decisions such as how to pass another vehicle/obstacle. Variational-based optimization techniques require more stringent restrictions on the vehicle's path to prevent collisions. The large scale path of the vehicle can be determined by using a voronoi diagram, an occupancy grid mapping, or a driving corridor algorithm. The latter allows the vehicle to locate and drive within open space that is bounded by lanes or barriers.

2.0 Content Block

Development General Motors is developing the "Ultra Cruise" ADAS system, that will be a dramatic improvement over their current "Super Cruise" system. Ultra Cruise will cover "95 percent" of driving scenarios on 2 million miles of roads in the US, according to the company. The system hardware in and around the car includes multiple cameras, short- and long-range radar, and a LiDAR sensor, and will be powered by the Qualcomm Snapdragon Ride Platform. The luxury Cadillac Celestiq electric vehicle will be one of the first vehicles to feature Ultra Cruise.

3.0 Content Block

Toyota In November 2022, Toyota demonstrated one of its GR Yaris test cars, which had been trained using professional rally drivers. Toyota used its collaboration with Microsoft in FIA World Rally Championship since the 2017 season.

4.0 Content Block

Maps Maps are necessary for navigation. Map sophistication varies from simple graphs that show which roads connect to each other, with details such as one-way vs two-way, to those that are highly detailed, with information about lanes, traffic controls, roadworks, and more. Researchers at the MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) developed a system called MapLite, which allows self-driving cars to drive with simple maps. The system combines the GPS position of the vehicle, a "sparse topological map" such as OpenStreetMap (which has only 2D road features), with sensors that observe road conditions. One issue with highly detailed maps is updating them as the world changes. Vehicles that can operate with less-detailed maps do not require frequent updates or geo-fencing.

5.0 Content Block

Driving systems Advanced driver-assistance systems (ADAS) automate specific driving features such as Forward Collision Warning (FCW), Automatic Emergency Braking (AEB), Lane Departure Warning (LDW), Lane Keeping Assistance (LKA) or Blind Spot Warning (BSW). An ADAS requires a human driver to handle tasks that the ADAS does not support. ADAS contrasts to an automated driving system (ADS), which would be classified by SAE J3016 as Level 3 or higher.

6.0 Content Block

Trolley problem The trolley problem is a thought experiment in ethics. Adapted for ACs, it considers an AC carrying one passenger confronting a pedestrian who steps in its way. The ADAS notionally has to choose between killing the pedestrian or swerving into a wall, killing the passenger. Possible frameworks include deontology (formal rules) and utilitarianism (harm reduction). One public opinion survey reported that harm

reduction was preferred, except that passengers wanted the vehicle to prefer them, while pedestrians took the opposite view. Utilitarian regulations were unpopular. Additionally, cultural viewpoints exert substantial influence on shaping responses to these ethical quandaries. Another study found that cultural biases impact preferences in prioritizing the rescue of certain individuals over others in car accident scenarios.

7.0 Content Block

Safety MRM A critical requirement for the higher two levels is that the vehicle be able to conduct a Minimum Risk Maneuver and stop safely out of traffic without driver intervention.

8.0 Content Block

Self-driving: "A vehicle 'satisfies the self-driving test' if it is designed or adapted with the intention that a feature of the vehicle will allow it to travel autonomously, and it is capable of doing so, by means of that feature, safely and legally." **Autonomy:** A vehicle travels "autonomously" if it is controlled by the vehicle, and neither the vehicle nor its surroundings are monitored by a person who can intervene. **Control:** control of vehicle motion. **Safe:** a vehicle that conforms to an acceptably safe standard. **Legal:** a vehicle that offers an acceptably low risk of committing a traffic infraction.

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Standards In April 2021, WP.29 GRVA proposed a "Test Method for Automated Driving (NATM)". In October 2021, Europe's pilot test, L3Pilot, demonstrated ADAS for cars in Hamburg, Germany, in conjunction with ITS World Congress 2021. SAE Level 3 and 4 functions were tested on ordinary roads. In November 2022, an International Standard ISO 34502 on "Scenario based safety evaluation framework" was published.

10.0 Content Block

Espionage In 2018 and 2019, former Apple engineers were charged with stealing information related to Apple's self-driving car project. In 2021 the United States Department of Justice (DOJ) accused Chinese security officials of coordinating a hacking campaign to steal information from government entities, including research related to autonomous vehicles. China has prepared "the Provisions on Management of Automotive Data Security (Trial) to protect its own data". Cellular Vehicle-to-Everything technologies are based on 5G wireless networks. As of November 2022, the US Congress was considering the possibility that imported Chinese AC technology could facilitate espionage. Testing of Chinese automated cars in the US has raised concern over which US data are collected by Chinese vehicles to be stored in China and any link with the Chinese communist party.

11.0 Content Block

Pedestrian detection In 2019, research from Georgia Tech revealed that autonomous vehicle detection systems were generally five percent less effective at recognizing darker-skinned individuals. This accuracy gap persisted despite adjustments for environmental variables like lighting and visual obstructions.

12.0 Content Block

Mobileye CEO Amnon Shashua and CTO Shai Shalev-Shwartz proposed an alternative taxonomy for autonomous driving systems, claiming that a more consumer-friendly approach was needed. Its categories reflect the amount of driver engagement that is required. Some vehicle makers have informally adopted some of the terminology involved, while not formally committing to it.

13.0 Content Block

Obstacles The primary obstacle to ACs is the advanced software and mapping required to make them work safely across the wide variety of conditions that drivers experience. In addition to handling day/night driving in good and bad weather on roads of arbitrary quality, ACs must cope with other vehicles, road

obstacles, poor/missing traffic controls, flawed maps, and handle endless edge cases, such as following the instructions of a police officer managing traffic at a crash site. Experimental studies show that trust in autonomous vehicles drops sharply after system failures, especially in control-related scenarios. Other obstacles include cost, liability, consumer reluctance, ethical dilemmas, security, privacy, and legal/regulatory framework. Further, AVs could automate the work of professional drivers, eliminating many jobs, which could slow acceptance.